

# wine\_analysis

May 4, 2025

## 0.0.1 Setup

To begin the project, I loaded the **red and white wine datasets** (from their CSV files in directory `datasets/`), which contain properties and quality scores for different wines.

I also used plot styling with Seaborn to ensure clean visuals and easy plot readability/interpretability throughout the notebook.

```
[15]: # Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load data
wine_red = pd.read_csv("../datasets/winequality-red.csv", sep=";")
wine_white = pd.read_csv("../datasets/winequality-white.csv", sep=";")
wine_red.columns = wine_red.columns.str.strip()
wine_white.columns = wine_white.columns.str.strip()

# Set global style
sns.set_theme(style="whitegrid", font_scale=1.2)
palette_red = sns.color_palette("Reds_r", as_cmap=True)
palette_white = sns.color_palette("Blues", as_cmap=True)
```

## 0.0.2 Task 1A: Wine Quality Comparison (Red vs White)

Below is a side-by-side plot comparing quality scores for both datasets. The white wine dataset (4,898 samples) is more than three times the size of the red wine dataset (1,599 samples) and therefore has a higher y-axis.

**Comparison:** - Both wines have median scores of around **5 and 6**. - **Red wine** is **highly focused** around these mid-scores, with almost no wines above score 7 (narrower spread). - **White wine** quality has a **wider range**, with more highly ranking wines (7–9) and a longer right tail (more dispersed).

```
[2]: # Plots

fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

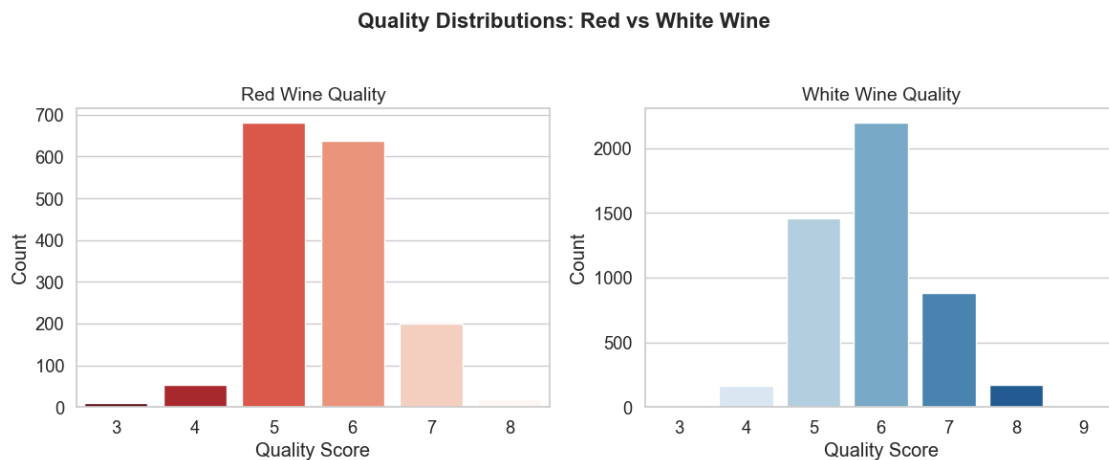
```

# Red plot
sns.countplot(data=wine_red, x="quality",
              hue="quality", ax=axes[0], palette="Reds_r", legend=False)
axes[0].set_title("Red Wine Quality", fontsize=14)
axes[0].set_xlabel("Quality Score")
axes[0].set_ylabel("Count")

# White plot
sns.countplot(data=wine_white,
              hue="quality", ax=axes[1], palette="Blues", legend=False)
axes[1].set_title("White Wine Quality", fontsize=14)
axes[1].set_xlabel("Quality Score")
axes[1].set_ylabel("Count")

# Both plot
plt.suptitle("Quality Distributions: Red vs White Wine", fontsize=16,
            weight="bold")
plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.savefig("../figures/wine_quality_comparison.png", dpi=300)
plt.show()

```



### 0.0.3 Red Wine Quality Distribution

This plot shows red wine quality ratings independently.

**Quality Plot Analysis:** - Most red wine qualities are rated either **5 or 6**, with not many wines outside of this range. - Few wines exist with quality scores of **3, 4, 7, or 8**, and there are none with a score of **9**. - The distribution is **unimodal** and suggests **low quality variance**.

```

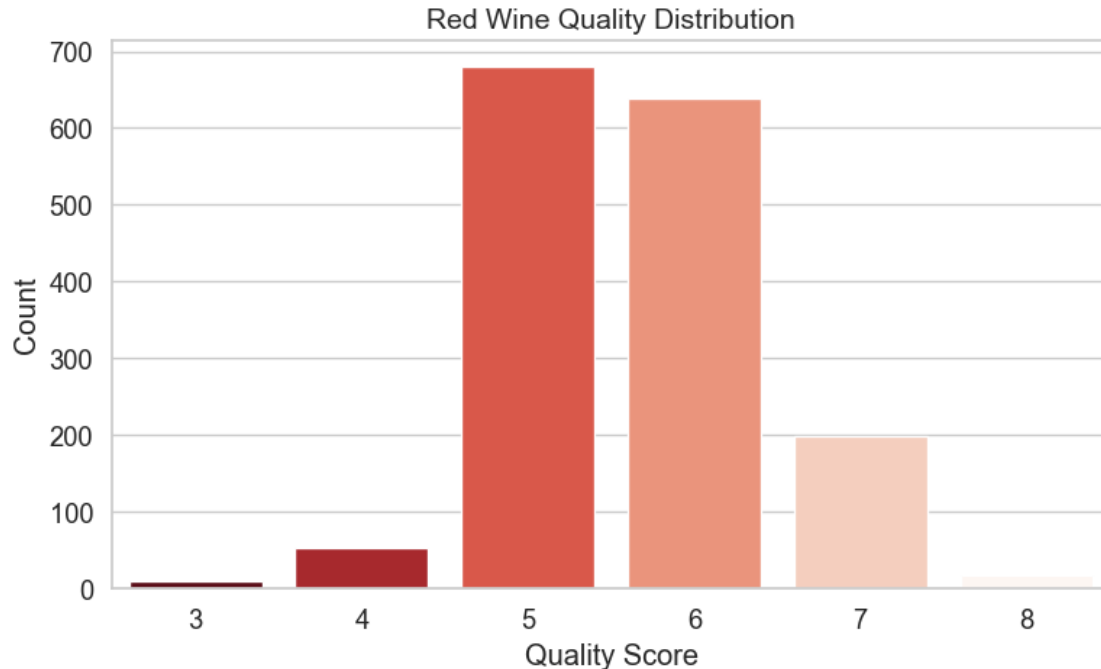
[3]: # Red wine plot
plt.figure(figsize=(8, 5))

```

```

sns.countplot(data=wine_red, x="quality", hue="quality", palette="Reds_r",
              legend=False)
plt.title("Red Wine Quality Distribution", fontsize=14)
plt.xlabel("Quality Score")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("../figures/red_wine_quality.png", dpi=300)
plt.show()

```



#### 0.0.4 White Wine Quality Distribution

The white wine dataset, when plotted against quality, is considerably different.

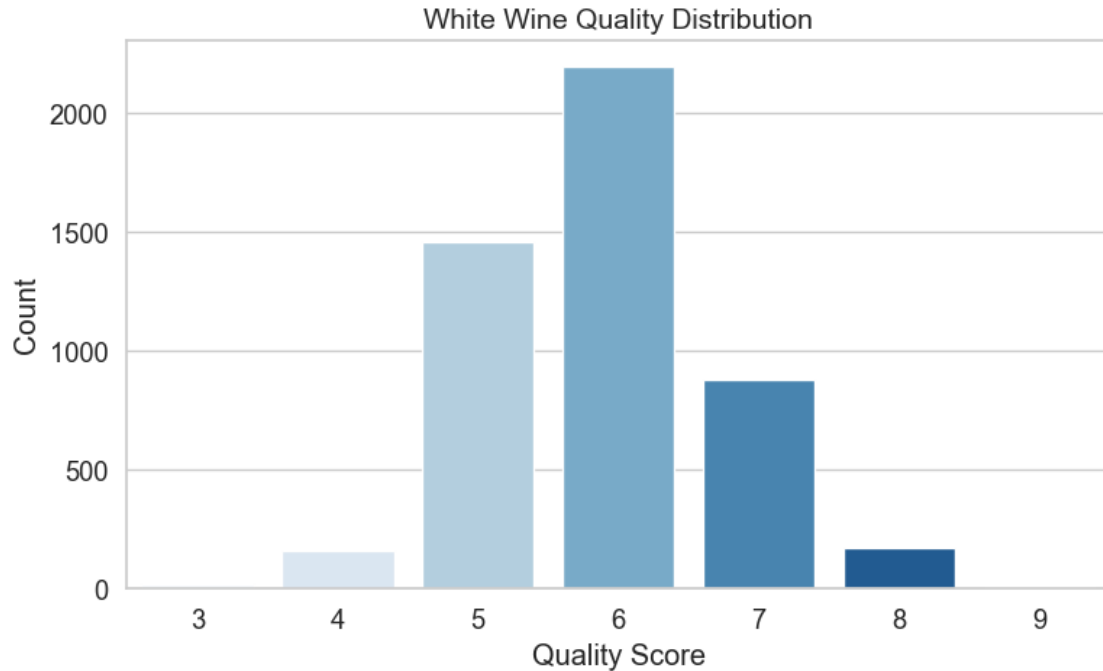
**Quality Plot Analysis:** - The mode is the same as red wine, at **6**, but the distribution is **broad** and includes higher ratings including **9**, unlike the red wine plot. - White wines rated **7 or 8** are not uncommon, and a few wines score at **9**. - Lower scores (**3, 4**) are also more frequent than in the red wines dataset.

```

[4]: # White wine plot
plt.figure(figsize=(8, 5))
sns.countplot(data=wine_white, x="quality", hue="quality", palette="Blues",
              legend=False)
plt.title("White Wine Quality Distribution", fontsize=14)
plt.xlabel("Quality Score")
plt.ylabel("Count")

```

```
plt.tight_layout()
plt.savefig("../figures/white_wine_quality.png", dpi=300)
plt.show()
```



## 0.1 End: Task 1A

### 0.1.1 Summary

- Red wine quality is tightly clustered around scores 5 and 6.
- White wine quality has a larger spread, with more highly-rated samples (7 and 8), and a longer right tail.
- There are over 3x as many white wine samples as there are red.

## 0.2 Begin Task 1B/1C

### 0.2.1 Discretise Alcohol Content Vars. Based on Distribution

To investigate how alcohol content influenced wine quality, I discretised alcohol content separately on whites and reds into a three valued variable (`alcohol_cat`) based on distribution:

- **low:**  $\text{alcohol} < (\text{mean} - \text{stddev})$
- **mid:**  $(\text{mean} - \text{stddev}) < \text{alcohol} < (\text{mean} + \text{stddev})$
- **high:**  $\text{alcohol} > (\text{mean} + \text{stddev})$

To do this, I made a helper function to calculate the dataset's mean and standard deviation (`stddev`), and assigned each sample to one of the three categories.

```
[18]: def add_alcohol_category(df):
    mean = df["alcohol"].mean()
    stddev = df["alcohol"].std()

    def categorise(alc):
        if alc < mean - stddev:
            return "low"
        elif alc > mean + stddev:
            return "high"
        else:
            return "mid"

    return df.assign(alcohol_cat=df["alcohol"].apply(categorise))
```

## 0.2.2 Generating Red Wine Quality Plot *by Alcohol Category*

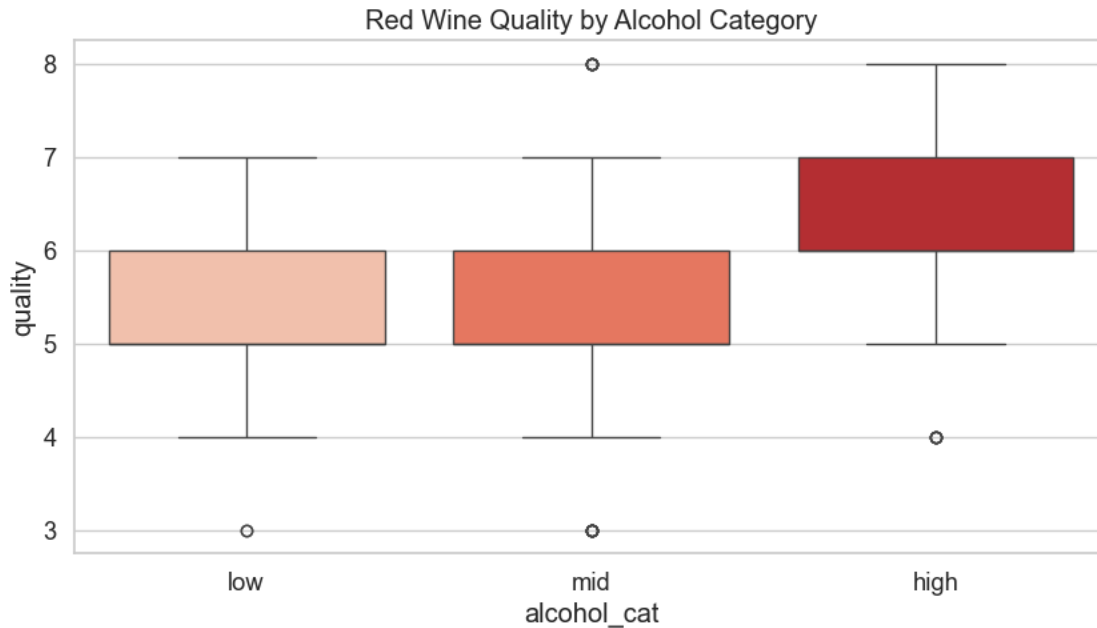
The boxplot below displays the relationship between red wine quality and `alcohol_cat` (low, mid, high).

**Observations:** - A clear positive trend is shown across alcohol categories: this suggests **higher alcohol content is associated with higher quality scores**. - The **median quality** increases from 5 (low) to 6 (mid), and 7 (high). - The **spread** of quality scores *narrows* for the high alcohol group, indicating that there is higher consistent quality among red wines with higher alcohol content. - Red wines in the low alcohol cat rarely achieve ratings above 6, and show more low outliers.

**Considerations:** - Alcohol seems to be influential to the quality of the red wines. - Thus the category `alcohol_cat` provides **meaningful stratification/classification** of quality and should be considered a possible feature in later predictive modelling.

```
[27]: # Apply to both datasets
wine_red = add_alcohol_category(wine_red)
wine_white = add_alcohol_category(wine_white)

# Red wine
plt.figure(figsize=(10, 5))
sns.boxplot(
    data=wine_red,
    x="alcohol_cat",
    y="quality",
    palette="Reds",
    hue="alcohol_cat",
    order=["low", "mid", "high"],
    hue_order=["low", "mid", "high"]
)
plt.title("Red Wine Quality by Alcohol Category")
plt.savefig("../figures/red_quality_by_alcohol.png")
plt.show()
```



### 0.2.3 Generating White Wine Quality Plot *by Alcohol Category*

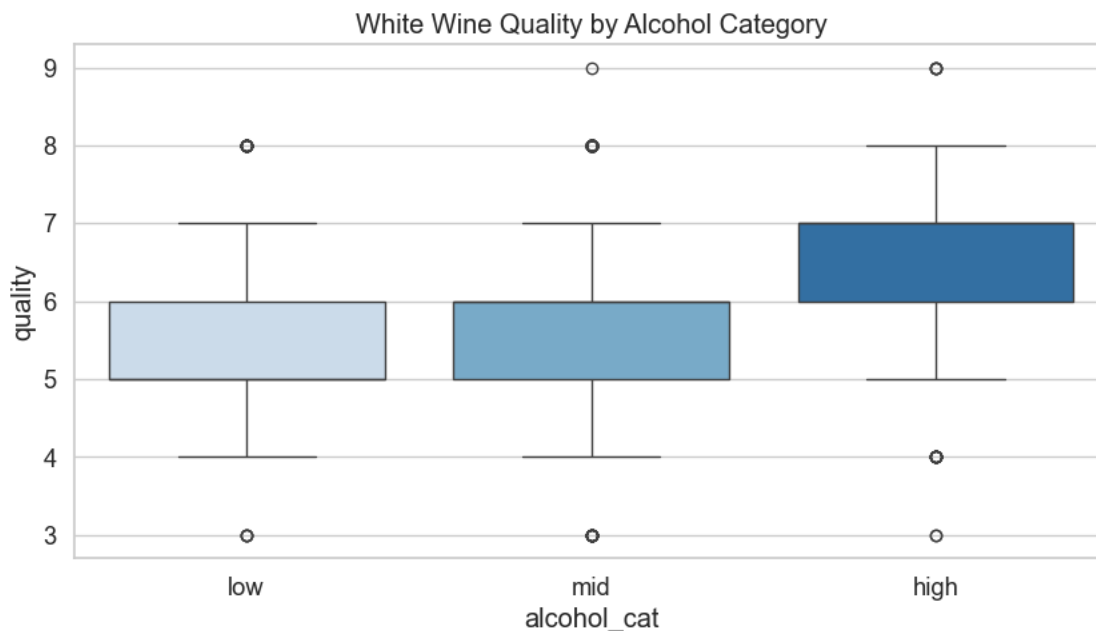
The boxplot below displays the relationship between white wine quality and `alcohol_cat` (low, mid, high).

**Observations:** - The same **positive correlation** seen in red wine also appears in white wine -> higher alcohol levels seem to map to higher quality. - The **high alcohol wines** have the highest median quality and includes most of the highest outlier quality wines (7–9). - Low alcohol wines are more tightly clustered around scores of 5–6, with less high-quality outliers. - Compared to red wine, the differences are slightly less dramatic, but still consistent.

**Considerations:** - Alcohol content is again **influential to quality**. - This consistent trend across both wine types suggests the `alcohol_cat` variable can be used influentially for both datasets, making it a valuable variable for model training across datasets later.

```
[56]: # White wine
plt.figure(figsize=(10, 5))
sns.boxplot(
    data=wine_white,
    x="alcohol_cat",
    y="quality",
    palette="Blues",
    hue="alcohol_cat",
    order=["low", "mid", "high"],
    hue_order=["low", "mid", "high"]
)
plt.title("White Wine Quality by Alcohol Category")
```

```
plt.savefig("../figures/white_quality_by_alcohol.png")
plt.show()
```



### 0.3 End: Task 1B/1C

#### 0.3.1 Summary

In Task 1B/1C, I plotted quality against alcohol content, based on distribution, and observed and analysed how it related to wine quality across both red and white wines.

To do this, I made a new variable, `alcohol_cat`, by discretising the `alcohol` variable into three levels (low, mid, high) using each dataset's mean and standard deviation.

#### Key Observations:

- For both red and white wines, **higher alcohol content correlates to higher quality**.
- The correlation is especially clear in red wines, where low alcohol wines rarely have higher quality ratings than 6, while high alcohol wines frequently have ratings of 7+.
- White wines show a similar, though less dramatic, trend — this means alcohol value can be used as a general predictor of quality.

#### For Later Modelling:

- The consistent correlation of alcohol content to quality suggests this feature can be **highly useful in both regression and classification models**.
- Using a discretised variable, such as `alcohol_cat`, instead of only alcohol values, also improves interpretability.

## 0.4 Begin Task 1D/1E

### 0.4.1 Plot Residual Sugar

The distribution of residual sugar is **skewed** for both red and white wine datasets, so to examine it's affect of quality, like with alcohol, I made a new binary variable **isSweet**.

**isSweet**, is: - 1 -> if residual sugar  $\geq$  median, - 0 -> else (so residual sugar will be  $\leq$  median).

Using the **median** means there will be an almost even split between sweet and dry, providing **balanced groups** with approximately the same amount of records.

```
[69]: def add_is_sweet(df):
        threshold = df["residual sugar"].median()
        return df.assign(isSweet=(df["residual sugar"] >= threshold).astype(int))

# Apply to both datasets
wine_red = add_is_sweet(wine_red)
wine_white = add_is_sweet(wine_white)
```

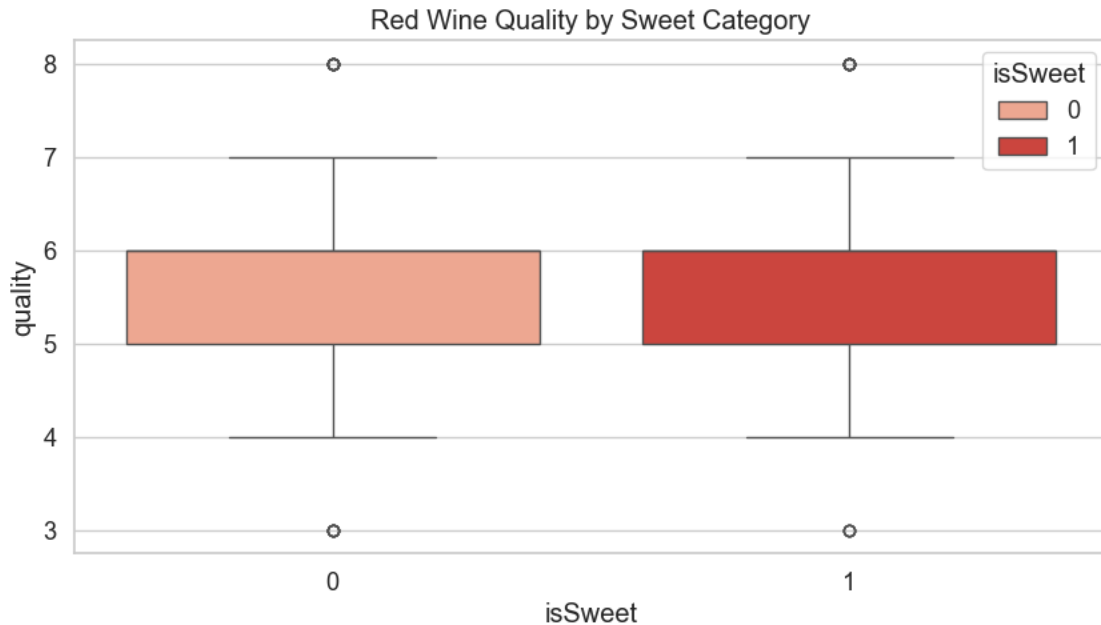
### 0.4.2 Red Wine: Quality by Sweetness

The boxplot below shows how quality varies between dry (**isSweet** = 0) and sweet (**isSweet** = 1) red wines.

**Observations:** - The **median quality is the same** for both groups (quality = 6). - Residual sugar *does not appear* to be influential in the quality of red wines. - This suggests that **sweetness alone is not a reliable indicator of quality** for red wine, though it may have different affect when combined and compared with other categories (like alcohol and sweetness).

```
[70]: # Red wine by sweetness
plt.figure(figsize=(10, 5))
sns.boxplot(data=wine_red, x="isSweet", y="quality", palette="Reds",
            hue="isSweet")
plt.title("Red Wine Quality by Sweet Category")
plt.savefig("../figures/red_quality_by_sugar.png")
plt.show()
```



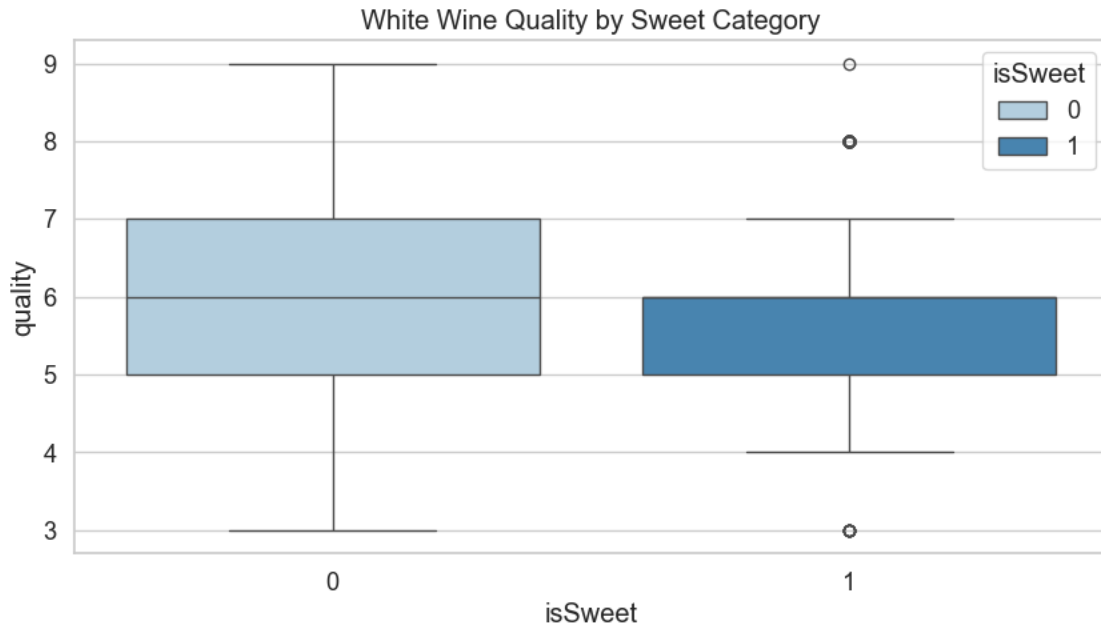


### 0.4.3 White Wine: Quality by Sweetness

The boxplot below shows how quality varies between dry (`isSweet = 0`) and sweet (`isSweet = 1`) white wines, and greatly differs from the red wine box plot.

**Observations:** - In contrast to red wine, dry white wines (`isSweet = 0`) show a **higher median** and have higher qualities. - Sweet white wines are much more tightly clustered and have less wines **exceeding a quality of 6**. - This indicates a clearer trend: **dry white wines are more likely to be rated highly**.

```
[71]: # White wine by sweetness
plt.figure(figsize=(10, 5))
sns.boxplot(data=wine_white, x="isSweet", y="quality", palette="Blues",
            hue="isSweet")
plt.title("White Wine Quality by Sweet Category")
plt.savefig("../figures/white_quality_by_sugar.png")
plt.show()
```



## 0.5 Summary: Task 1D/1E Findings

### 0.5.1 Overall:

- Created an `isSweet` binary variable based on the median residual sugar for each wine type to guarantee an even split between dry and sweet wines in each dataset.
- Sweetness shows **little correlation with quality** in red wine, but a **clear negative trend** in white wine.

### 0.5.2 For Later Modelling:

- `isSweet` may not be a useful indicator for red wine prediction.
- It may improve classification performance for white wines when used in combination with other categories.

## 0.6 Begin Task 2

### 0.6.1 Correlation Matrix:

To determine which subset of variables would be most useful for learning, I made a correlation matrix to analyse relationships between: - Each pair of variables - Each variable and the outcome (quality) - Plotted visually

I used the **Pearson correlation**, which measures **linear relationships** between continuous variables. I chose this metric because: - All variables are numerical - Most relationships with quality are linear

## 0.6.2 Predictions:

Based on the previous plots, I predict: - Alcohol content will be a strong indicator of quality - Higher sweetness will negatively affect white wines quality

## 0.6.3 Red Wine: Correlation Matrix

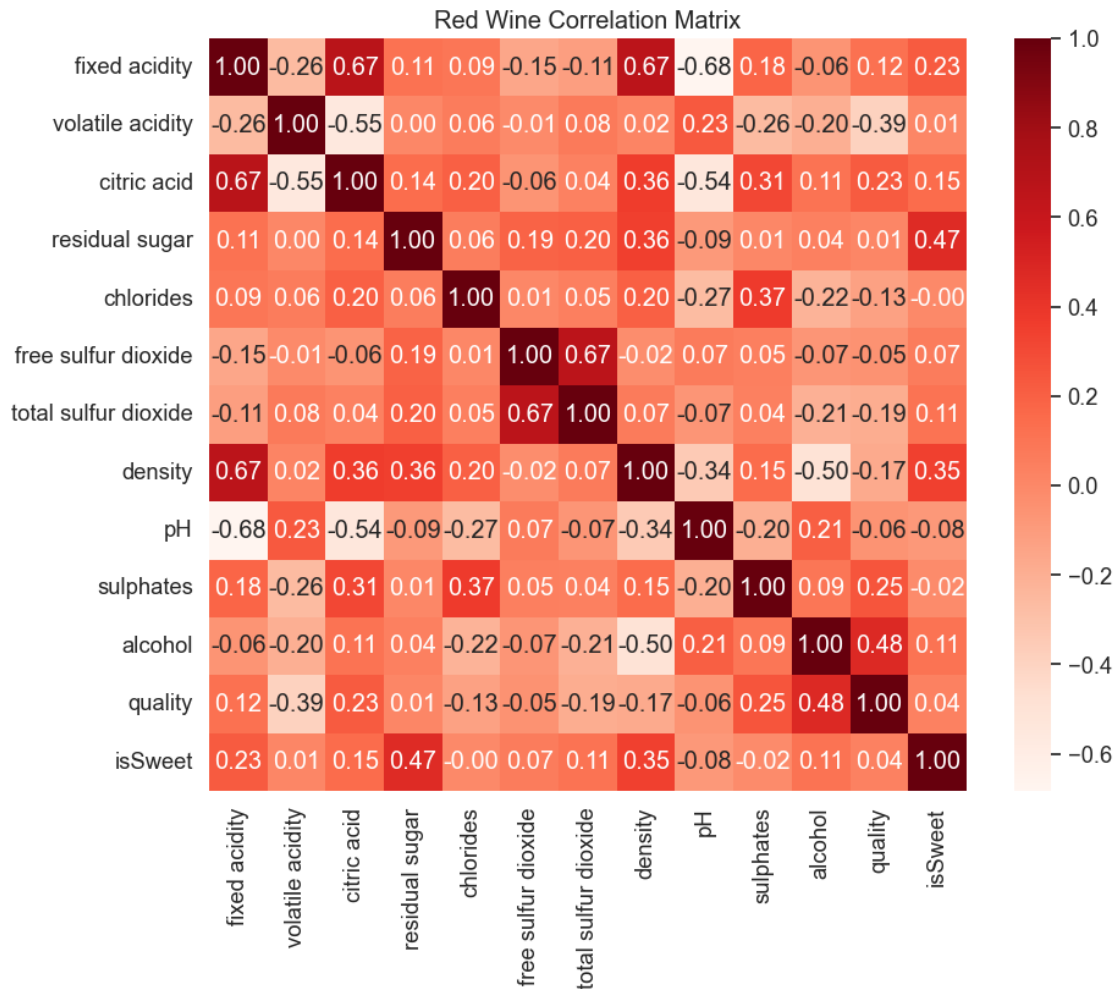
The matrix below shows how strongly each red wine variable correlates with others.

### Key Observations:

- **Alcohol** has a somewhat **positive correlation** with quality (0.48), suggesting higher alcohol content typically means the red wines quality is higher.
- **Volatile acidity** has a **negative correlation** (-0.39), which suggests higher acidity often corresponds to **lower quality**.
- **Sulphates** and **citric acid** also show positive correlations with quality.
- **Fixed acidity** and **citric acid** are positively correlated with each other — having both as indicators may be redundant.
- **Free and total sulphur dioxide** are strongly correlated with each other (0.67), so may also be redundant.

```
[74]: # Define correlation matrix using pearson for both wines
corr_red = wine_red.corr(numeric_only=True, method='pearson')
corr_white = wine_white.corr(numeric_only=True, method='pearson')

# Red wine correlation matrix
plt.figure(figsize=(10, 8))
sns.heatmap(corr_red, annot=True, cmap="Reds", fmt=".2f")
plt.title("Red Wine Correlation Matrix")
plt.savefig("../figures/red_wine_matrix.png")
plt.show()
```



#### 0.6.4 White Wine: Correlation Matrix

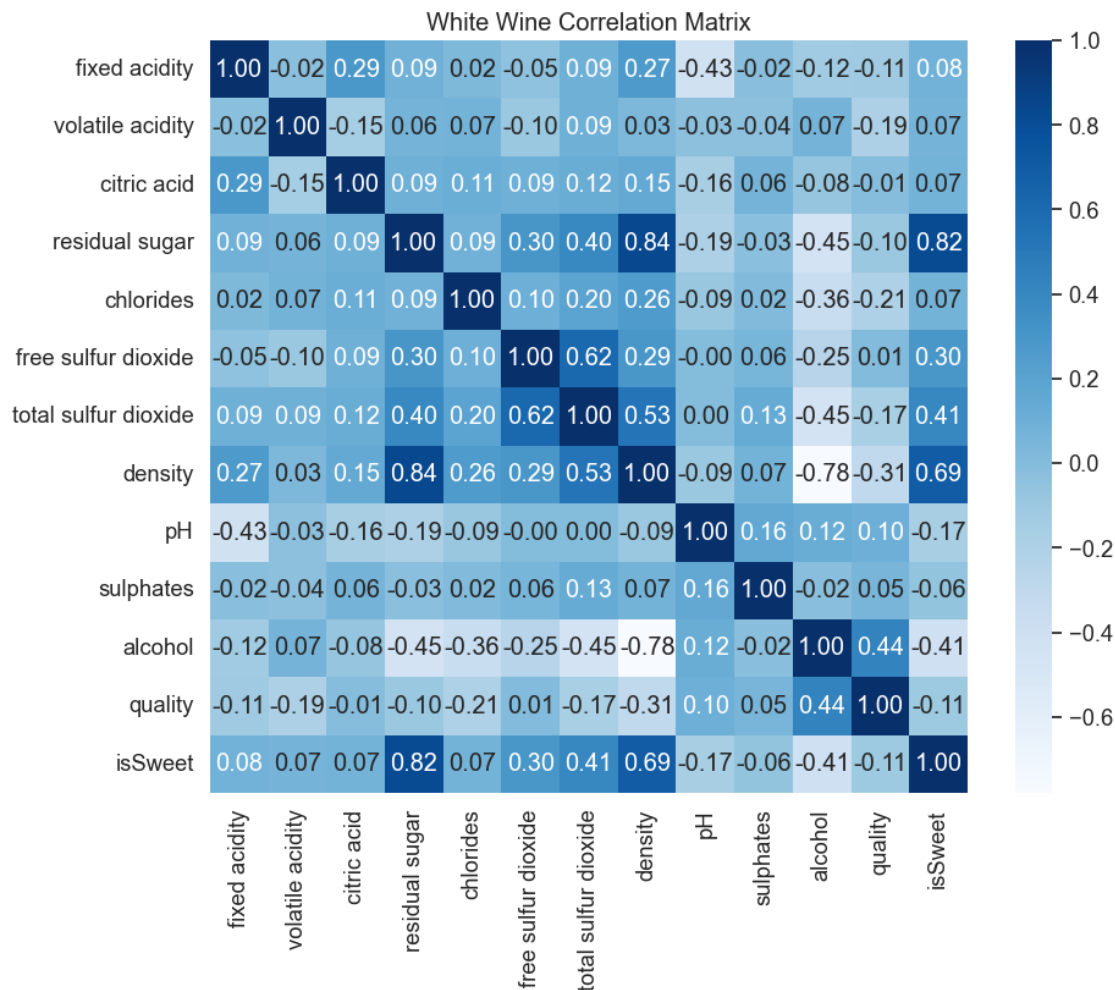
The matrix below shows how strongly each white wine variable correlates with others.

##### Key Observations:

- **Alcohol** again shows a **positive correlation** with quality (0.44), consistent with red wines.
- **Residual sugar** and **density** show **negative correlation** with quality.
- **Free sulphur dioxide** and **total sulphur dioxide** are **highly correlated** (0.72), similar to the red wine dataset, meaning having both here could be redundant as well.
- **Chlorides** and **density** are moderately correlated — could also be considered redundant.

```
[75]: # White wine correlation matrix
plt.figure(figsize=(10, 8))
sns.heatmap(corr_white, annot=True, cmap="Blues", fmt=".2f")
plt.title("White Wine Correlation Matrix")
plt.savefig("../figures/white_wine_matrix.png")
```

```
plt.show()
```



## 0.7 Task 2 Summary:

From both correlation matrix's, **alcohol** stands out as the best **positive predictor** of wine quality, while **volatile acidity** (*for red*) and **density** (*for white*) could be used as **possible negative predictors** in quality.

### For Later Modelling:

- Use *alcohol*, *sulphates*, *citric acid*, and *pH*.
- Don't use one of **free sulphur dioxide** or **total sulphur dioxide** as matrixs suggests them to be so closely correlated one is redundant.
- **Residual sugar** may be important indirectly (as seen in sweetness analysis), but direct correlation with quality is low.
- Some may be useful only in **non linear** models, despite weak Pearson correlation (as Pearson focus on linear relationships).